

Python Full-Stack Web Application for Data Management

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ABSTRACT

Modern web applications require seamless integration between front-end and back-end technologies to provide efficient and user-friendly systems. Traditional standalone applications lack scalability, real-time interaction, and centralized data management.

This paper presents the design and implementation of a Python-based full-stack web application that integrates front-end technologies (HTML, CSS, JavaScript) with a Python back-end framework (Flask/Django) and a relational database system. The system provides secure user authentication, efficient data management using CRUD operations, and a responsive user interface.

Keywords: Python, Full-Stack Development, Flask, Web Application, CRUD Operations, User Authentication

1. INTRODUCTION

Full-stack development combines front-end and back-end technologies...

2. PROPOSED SYSTEM

User Module, Backend, Database Layer explained...

Fig 1. System Architecture (Figure Required)

3. IMPLEMENTATION AND RESULT

System tested with login, CRUD operations, responsive UI.

Fig 2. Login Page (Figure Required)

Fig 3. Dashboard (Figure Required)

5. CONCLUSION

The project demonstrates integration of technologies and serves as a foundation.

6. REFERENCES

Flask, Django, W3Schools, SQLite, MDN